Energy Strategy Reviews 56 (2024) 101583

Contents lists available at ScienceDirect

Energy Strategy Reviews

journal homepage: www.elsevier.com/locate/esr

Regional digital infrastructure and carbon neutrality: A technology–structure–efficiency perspective

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ARTICLE INFO

Handling editor: Mark Howells

Keywords:
Carbon neutrality
Digital infrastructure
Sustainability
Broadband China
Industrial structure

ABSTRACT

The synergistic development of digital transformation and carbon neutrality is a key path to global sustainable development. This study utilized the Broadband China pilot policy and a multi-period Difference-in-Differences (DID) design to comprehensively evaluate the impact of digital infrastructure on carbon neutrality using panel data from 291 prefecture-level cities in China from 2008 to 2021. The results indicate that regional digital infrastructure contributed to carbon neutralization innovation, carbon-intensive industry innovation, and carbon-emission efficiency in the pilot areas. This held true after analyzing the results with propensity score matching, coarsened exact matching, robust estimators, and instrumental variable regression. Specifically, the results of the mediation effect model indicate that digital infrastructure promotes the process of carbon neutrality through several mechanisms, including promoting the rationalization of industrial structure, promoting digital economic development, and inducing innovation in digital technology. In addition, cities with better economic development, lower resource dependence, and weaker environmental constraints have a greater carbon neutrality dividend effect from regional digital infrastructure. These findings indicate a need to optimize policies to unleash the dividends of digital infrastructure in carbon neutrality, which can drive technological, structural, and efficiency changes through digital empowerment. In addition, cities with heterogeneous endowments need to develop targeted measures.

1. Introduction

Global carbon emissions in industrial societies have contributed to climate change, resulting in devastating environmental and social disasters [1]. Adverse climate change not only directly causes agricultural production crises but also affects the sustainable development of almost all industries [2]. The increase in extreme weather events has caused unbearable losses to human society, and various countries have developed measures to reduce greenhouse gas emissions [3]. As one of the world's largest economies, China has shown great concern for global carbon control and has taken initiatives to reach a carbon peak by 2030 and carbon neutrality by 2060. The upcoming digital era might provide opportunities for carbon neutralization, and digital infrastructure seems to provide a new impetus for economic decarbonization, which can further facilitate the achievement of sustainable growth. Thus, confirming the potential impact and acting mechanisms of digital infrastructure on carbon neutrality is likely to illuminate the future path of green development.

The Broadband China pilot policy is a major attempt to realize digitally driven sustainable development in China. It offers fiscal priority and technical support for nationwide broadband construction. The Chinese national council announced three batches of Broadband China pilot cities in 2014, 2015, and 2016. The internet penetration rate reached 55.8 % by the end of 2017 in pilot cities, and the number of internet broadband access ports made a net increase of 71.66 million within a year. Digital infrastructure has significantly benefited from the Broadband China pilot policy, which can therefore serve as an intervention experiment for changes in regional digital infrastructure. Digital infrastructure enhances the potential for the digital transformation of the economy and society and provides opportunities for the transformation of the energy system and the development of renewable energy systems [4]. Studies have shown that digital infrastructure can facilitate the operation of smart energy systems [5] and support technological innovation for carbon reduction [6]. Digital infrastructure also supports the development of low-carbon industries, which in turn may influence the process of carbon neutrality through various mechanisms.

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https://doi.org/10.1016/j.esr.2024.101583
Received 27 June 2024; Received in revised form 23 October 2024; Accepted 2 November 2024

The main source of carbon emissions is the consumption of fossil energy sources, and the first step to promoting carbon neutrality is to reduce both energy intensity and dependence on such sources [7]. Energy-related technological innovations drive improved energy efficiency and carbon-disposal efficiency [8,9], while this process remains seamless and efficient with energy information flows [10]. In addition, green technology innovation has a significant impact on decarbonization, of which carbon-neutral technology innovation is a key driver [11]. According to Martínez et al. [12], infrastructure-based digitalization is vital for green development and sustainable growth and is a key component of smart energy systems. In sum, the digitalization of the economy leads to increased efficiency and innovation [13], creating new opportunities for achieving carbon neutrality.

Using the panel data of 291 Chinese prefecture-level cities, we calculates carbon-emission efficiency using the data envelopment analysis (DEA) method, carbon-intensive industries innovation, and carbon-neutral technological innovation to evaluate progress toward carbon neutrality from a comprehensive perspective. The study uses the Difference-in-Differences (DID) method to determine whether digital infrastructure can significantly promote carbon neutralization. It also uses industrial structure, the digital economy, and digital innovation as mediating variables to investigate the mechanisms. This study demonstrates the promoting effect of digital infrastructure on carbon neutralization and applies the standard deviation of altitude as an instrumental variable. Furthermore, it shows that digital infrastructure promotes carbon neutrality through industrial structure rationalization, digital economy development, and digital innovation. Finally, this study empirically discusses the heterogeneity effects caused by urban endowments and concludes that the carbon-neutral effects of digital infrastructure require financial support from local governments to escape the resource dependence trap.

This study makes the following marginal contributions: It describes carbon neutrality from the perspective of technical structural efficiency, deconstructs the key components of carbon neutrality, and provides a perspective for understanding the evolution law of carbon neutrality in the age of digital intelligence. It also provides a theoretical framework for the synergistic development of digital transformation and carbon neutralization. Within the framework of digital transformation, this study elucidates the transmission mechanism of the carbon-neutral effects of the digital economy and digital technology innovation in regional digital infrastructure. It also provides empirical inspiration and differentiated policy guidance for cities with heterogeneous endowments to achieve smart carbon neutrality.

The remaining sections are arranged as follows: Section 2 presents the theoretical analysis and research hypotheses. Section 3 describes the research design, variables, and models. Section 4 presents the results and analysis of the benchmark regression, robustness tests, mechanism tests, and heterogeneity tests. Section 5 provides policy recommendations and research prospects. The analytical framework of this study is shown in Fig. 1.

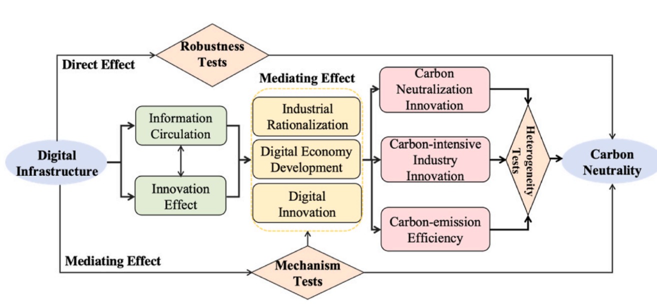

Fig. 1. Analytical framework.

2. Literature review and theoretical analysis

2.1. Literature review

The following possible approaches to carbon neutrality have been explored in much of the literature, The path to decarbonizing the economy, which requires shifting economic growth away from heavy dependence on fossil fuels, has been widely discussed [7]. Both carbon removal technologies [14] and carbon absorption technologies [8] provide technical support for achieving the goal of carbon neutrality. Aside from technological innovation, new policies are also beneficial. Carbon pricing has been shown to be effective in rationalizing carbon emissions [15]. Other market-based systems for emission management, such as emission permit trading, have already gained attention in China and other countries [16]; emission trading schemes match emissions and efficiency better and encourage green technological innovation in industrial enterprises [17]. Green fiscal supports aimed at regional sustainable development also have an impact on diminishing the level and intensity of carbon emissions [18]. Finland's avantgarde experience revealed other key points about carbon neutrality, such as integrated governance, intellectual property barriers, and data-related issues [10], all of which require highly efficient information circulation and strong innovative capability.

Digital infrastructure has certain characteristics that relate to carbon neutralization. It lays the groundwork for green development: network and information infrastructure serve as the material underpinning for the digital economy, hatching new businesses and platforms for digital transformation [4], which is a key component of green development and sustainable growth [12,19]. The application of information and communication technology allows for more efficient information circulation, which helps optimize production division and specialization [20], ultimately reducing reliance on fossil fuels. Novel applications derived from digital infrastructure, such as artificial intelligence, reduce energy intensity by promoting the digital transformation of micro-economic units [21]. More direct evidence has confirmed that it restricts carbon emissions in Chinese provinces and facilitates the invention of new technologies for emission control [22]. It also helps ease energy intensity and has an impact on energy structure [23].

Digital infrastructure makes innovation activities less capital intensive via knowledge penetration [13]. It also strengthens digital inclusive finance [24] and green finance [25], while these new financial tools lubricate sustainable transformation. Internet use and digital integration affect the sustainable competitiveness and entrepreneurial vitality of a country to varying degrees [26]. Research has shown that digital infrastructure improves enterprise performance by reducing external costs and strengthening internal control [27]. These performance improvements ultimately contribute to increasing an economy's green total factor productivity [6]. Broadband networks can improve enterprise production efficiency [28], as exemplified by the Smart Grid Program in Canada [29], which found that digital network infrastructure improved the efficiency of the grid through real-time information feedback channels and reduced energy waste and overall emissions.

Most of the literature is devoted to a broad theoretical framework describing the interaction between carbon neutrality and digital transformation. While there is a clear theoretical correlation between digital infrastructure and carbon neutrality, it has only appeared in previous work as a supplementary discussion; a direct empirical investigation of this relationship should be valuable. Instead of focusing on isolated components of carbon neutrality, this study describes carbon neutrality as a synthesis of carbon-emission efficiency, carbon-intensive industry innovation, and carbon neutralization innovation.

2.2. Theoretical analysis of direct effects

Carbon neutrality consists of three dimensions—low carbon, zero carbon, and negative carbon—which correspond to energy efficiency improvement, economic decarbonization, and carbon disposal techniques, respectively. Comprehensive application of digital infrastructure reshapes the industrial form and development method, reduces dependence on fossil energy, optimizes energy consumption, and provides new opportunities for low-carbon development of the global economy.

First, digital infrastructure improves energy efficiency and reduces carbon emissions. It helps reduce production costs and enhance production efficiency by providing a smoother information circulation mechanism and technological effects [4], closing the actual decision-making curve with the efficiency frontier, enlarging expected output per unit of energy consumption, and reducing emissions under the same output. Additionally, the application of digital infrastructure lays the foundation for advanced novel industries, thus promoting the optimization of industrial structure [30]. This will further adjust the current energy consumption structure, reduce the intensity of fossil energy dependence [31], and, in turn, reduce emissions.

Second, digital infrastructure facilitates innovation in carbon-intensive industries and promotes the clean transformation of traditionally high-emissions industries. Digital infrastructure has proven to be effective in guiding production servitization and industrial innovation, which provides an impetus for transformation in carbon-intensive industries [32]. In addition, digital infrastructure helps cut production and management costs, thereby retaining internal financial resources for technological innovation. It also promotes the further development of green finance, thus providing external credit support for green innovation [33]. Green innovation in carbon-intensive industries can provide technical support for clean transformation. The rise of clean industries completes the zero-carbon node in carbon neutralization.

Third, digital infrastructure drives carbon-neutral technological innovation. Digital infrastructure makes it possible to collect, analyze, and apply environmental information on an enormous scale, thus providing support for carbon-emission monitoring, carbon capture, and climate change adaptation [34], and invents new application scenarios for carbon-neutral related technologies. Hence, digital infrastructure completes the carbon neutralization process of low-carbon, zero-carbon, and negative-carbon. Based on the above propositions, the following hypothesis was developed:

Hypothesis 1. Regional digital infrastructure can significantly contribute to the realization of carbon neutrality.

2.3. Theoretical analysis of mediating effects

Digital infrastructure promotes carbon neutrality through industrial structure rationalization. Digital industries develop on the basis of infrastructure, provide alternative destinations for factor flow, capital, and labor are more likely to achieve higher efficiency in these novel industries, reallocated factor, and better matched production rate, thus rationalizing the industrial structure [35]. When facing the constant allocation of factors, digital finance and information-circulation tools can alleviate the disruption of vulnerable industries and small enterprises [24] and thus lubricate the rationalization of industrial structure. This means production factors flow to the industrial sector with the highest relative utilization efficiency, thus enhancing the overall production efficiency at a fixed technological level and ultimately leading to the enhancement of output per unit of carbon emissions. This manifests in the reduction of total resource consumption and resource waste driven by the improvement of utilization efficiency [36]. Meanwhile, the improvement of overall production efficiency can provide an intrinsic driving force for further innovation, which contributes to carbon-neutrality-related technological innovation. Moreover, research has shown that intelligent manufacturing promotes industrial upgrading that can facilitate low-carbon transition [37], which is a clear demonstration of industrial structure's mediating role in the carbon-neutral effect of digital infrastructure. Hence, the following hypothesis was developed:

Hypothesis 2. Digital infrastructure promotes carbon neutralization via industrial structure rationalization.

Digital infrastructure leads to carbon neutrality through the digital economy. Digital infrastructure primarily provides a material foundation for the digital transformation of existing industries, and the synergistic development digitalization will also incubate novel digital industries, which are the main components of the digital economy. Thus, the development of digital infrastructure can promote the prosperity of the digital economy. Meanwhile, the digital economy is the key to achieving carbon neutrality. Economic digital transformation can digitalize physical production and management processes, reducing physical resource consumption and pollution emissions [38]. Under the circumstance of low-carbon construction, the digital economy has been shown to enhance green development by adjusting industrial structure and boosting innovation [39]. In addition, the digital economy promotes the development of clean energy [40] and green total factor productivity [41]. The widespread application of the digital economy can also improve governance [42] and realize comprehensive green development. Therefore, the following hypothesis was developed:

Hypothesis 3. Digital infrastructure promotes carbon neutralization through digital economy development.

Digital infrastructure helps carbon neutralization via digital innovation. Infrastructure-driven digitalization can smooth internal and external information circulation and utilization, resulting in lower production and management costs, which can relieve the financial restraint of innovation entities [43]. Digital technologies also provide crucial information to support the interactions and integration of different innovation sectors, hence constructing a quasi-ecosystem as a new organizational form for all kinds of innovation actors [44]. Once embedded in such an ecosystem, complementary technological linkages between different entities begin to take effect, and an inverted U-shaped curve relationship between this linkage and carbon-emission efficiency is observed [45]. This innovative effect has spatial spillovers [46]. Digital innovation is a powerful engine for carbon neutralization. It influences the affordability of digital technologies to promote innovation efficiency and economic efficiency and then supports emission reductions in the carbon neutralization process [27]. Digital innovation itself also brings new technologies to support improvements in energy and waste disposal efficiency to provide a new impetus for carbon neutralization. This leads to higher efficiency and more advanced green technology, which bring carbon neutrality closer. Hence, the following hypothesis was developed:

Hypothesis 4. Digital infrastructure promotes carbon neutralization by improving digital innovation.

3. Materials and methods

3.1. Sample selection and data sources

Relevant data from 291 prefecture-level cities in 30 Chinese provinces were used as samples, with an observation period ranging from 2008 to 2021. The author manually compiled the list of Broadband China pilot cities based on the State Council's Notice on Issuing the Broadband China Strategy and Implementation Plan, as well as the list provided by China's Ministry of Industry and Information Technology. The China Energy Statistical Yearbook provides input-output data (working population, energy consumption, total output value, and carbon dioxide emissions) for calculating carbon-emission efficiency (*CEE*).

The source data for the carbon-intensive industry innovation index (*CII*) is from the Report on Chinese Cities and Industrial Innovation Power released by the Industrial Development Research Center of Fudan University. The carbon neutralization innovation index (*CNI*) is the sum of the number of patent applications in carbon-neutrality-related fields, with data obtained from the IncoPat Global Patent Database. Among the mediating variables, the data for the digital economy development index were obtained from the China City Statistical Yearbook; the patent data for digital economy innovation was also obtained from the Incopat patent database. The altitude standard deviation data used as instrumental variables were obtained from the geospatial data cloud platform of the Computer Network Information Center (Chinese Academy of Sciences, http://www.gscloud.cn). The data for the remaining control variables were obtained from the China City Statistical Yearbook.

3.2. Definitions of variables

Three dependent variables were used to construct a multidimensional index system in this paper. *CEE* was calculated using the Data Envelopment Analysis (DEA) method, taking the number of industrial employees, foreign investment, and energy consumption as input factors; gross domestic product (GDP) as expected output; and carbon dioxide as unexpected output.

DEA uses the input and output variables, as shown in Table 1, to construct decision-making units (DMUs). The efficiency of each DMU is calculated using linear programming. Efficiency is one of these efficient units, and inputs and outputs of efficient DMUs are used to construct the efficiency frontier and to measure the relative efficiency of other DMUs. The smaller the distance from the efficiency frontier, the higher the efficiency. The Slacks-Based Measure Data Envelopment Analysis (SBM-DEA) model introduces slack variables, which can measure input redundancy and output shortfalls more accurately than the proportional input/output evaluation of traditional DEA while also taking unexpected outputs into consideration. In the context of this study, SBM-DEA calculated the input/output efficiencies of labor, capital, and energy inputs, as well as expected GDP, and unexpected CO_2 for each city, indicating the efficiency of economic development under the constraints of carbon-emission reduction. In this study, it was used as an expression of carbon-emission reduction in the carbon neutralization process.

The *CII* comes from the Industrial Development Research Center of Fudan University, and the annual stock of patent value was readjusted through the patent update model to obtain the innovation index of each industry. The innovation of carbon-intensive industries was constructed according to the method of Liu and Zhao (2017), combining the innovation index of related industries. The CNI refers to the number of patent applications representing technological innovations in areas related to carbon neutrality. According to the standards of the European patent office database, eight patent projects within the CPC-Y02 category under the international patent classification number system were identified as carbon neutrality fields, including climate change adaptation technology, climate change mitigation technology, greenhouse gas disposal and reduction technology, and waste disposal technology.

The core explanatory variable (*did*) was the dummy variable of the Broadband China pilot policy, which consisted of a group dummy variable (*treat*) and a year dummy variable (*post*). This *did* was given a value of 1 only when the city was in the pilot list after the policy was implemented and 0 otherwise. Broadband China is a large-scale internet infrastructure construction. Broadband is one of the main components of digital infrastructure, and the internet is the basic platform for digital transformation.

Table 1
Calculation of *CEE*.

Types	Variables	Definitions	Sources
Input	Labor	Sum of second and third industries' employee	China City Statistical Yearbook
	Investment	Sum of foreign investment actually used in the current year	China City Statistical Yearbook
	Energy	Sum of cities' electricity, coal, and natural gas consumption	China Energy Statistical Yearbook
Expected output	GDP	Each city's gross domestic product	China City Statistical Yearbook
Unexpected output	CO_2	Each cities' annual carbon dioxide emission	China Energy Statistical Yearbook

This study used a control variable set that mainly covered the following: Proportion of third industry (*Trd_Ind*) and proportion of employees in third industry (*Trd_Ind*) represent the industrial structure of a city and pattern of a city comprehensively. Sulfur dioxide emission (SO_2) is another main emission in industrial production other than carbon dioxide; it reflects the energy consumption and green development of the city. Foreign direct investment (*FDI*) partly represents the transfer of energy consumption and environmental pollution according to the pollution haven hypothesis. Since China is one of the world's major host countries for international investment, it is necessary to take the impact of foreign investment on carbon neutrality into account. Natural gas consumption (*Gas*) directly reflects a city's energy utilization intensity and indirectly reflects its energy structure, both of which significantly influence the carbon neutrality process. Permanent population (*Pop*) and gross domestic production growth rate (*GDPr*) reflect a city's economic development, which is the premise factor of a low-carbon economy. Carbon dioxide emission (CO_2) indicates urban energy consumption, industrial production intensity, and energy dependence, making it a core index of carbon neutrality and green economy issues. Fiscal revenue (*Gov_In*) and expenditure (*Gov_Out*) represent the power and willingness of local governments to intervene in carbon neutralization and have an important influence on *CNI*. Fiscal expenditure on education (*Edu_Out)* is also related to regional innovation capacity, which indirectly affects industrial structure.

Industrial rationalization (*RIS*), digital economy development (*Digit_Eco*), and digital innovation (*Digit_Pat*) were selected as mediating variables. Following Gao et al. [35], this study used the structural deviation method and Hamming closeness evaluation method to construct the rationalization indexes of the second and the third industries using Eq. (1):

$$RIS_{i,t} = 1 - \frac{1}{2}\sum_{n=1}^{2}\left|\frac{Ind_{i,n,t}}{GDP_{i,t}} - \frac{Ind_{Pop_{i,n,t}}}{Labor_{i,t}}\right| \tag{1}$$

where $\frac{Ind_{i,n,t}}{GDP_{i,t}}$ indicates the proportion of output from a certain industry in the total output in year t and $\frac{Ind\_Pop_{i,n,t}}{Labor_{i,t}}$ is the proportion of industry employees in the total labor population in year t. The larger the *RIS*, the more reasonable the industrial structure. While the rationalization of industrial structure is a reflection of coordinated development among industries, it also reflects the efficiency of the distribution and utilization of resources among industrial sectors.

This study also adopted the digital economy development index (*Digit_Eco*), which comprises the internet penetration rate, internet-related output, digital inclusive finance and other elements, and the logarithm of digital-economy-related patents (*Digit_Pat*) as mediating variables. The identification of patents related to the digital economy was conducted in accordance with the Statistical Classification of Digital Economy and Its Core Industries (2021) issued by the National Bureau of Statistics, covering digital product manufacturing industries, digital product service industries, digital technology application industries, and digital factor-driven industries. Table 2 shows the definitions of relevant variables in this study

3.3. Descriptive statistics

Descriptive statistics for each variable are presented in Table 3. To ensure the simplicity of the regression results, the logarithms of all continuous variables were taken, samples with excessive missing variables were removed, and variables with only countable missing were

Table 2
Variable definitions.

Types	Variables	Symbols	Definitions	Unit
Policy	Digital infrastructure	did	Dummy variable for the pilot policy	–
Dependent Variables	Carbon-emission efficiency	*CEE*	SBM-DEA algorithm using carbon dioxide as unexpected output	–
	Carbon-intensive industries innovation	*CII*	Annual readjusted stock patent value in carbon intensive industries	–
	Carbon neutralization innovation	*CNI*	Logarithm of annual patent application in carbon-neutral fields	1,000 pieces
Control Variables	Industrial structure	*Trd_Ind*	Proportion of gross production of the third industry	%
		Trd_Pop	Proportion of employees of the third industry	%
	Greenhouse gases	SO_2	Logarithm of sulfur dioxide emission	10,000 tons
		CO_2	Logarithm of carbon dioxide emission	10,000 tons
	Energy consumption	*Gas*	Logarithm of natural gas consumption	10,000 m^3
	Energy demand factors	*GDPr*	Annual growth rate of gross domestic production	%
		FDI	Logarithm of foreign direct investment	10,000 dollars
		Pop	Logarithm of permanent population	10,000 people
		Gov_Out	Logarithm of government fiscal expenditure	10,000 yuan
	Government interventions	*Gov_In*	Logarithm of government fiscal revenue	10,000 yuan
		Edu_Out	Logarithm of government educational and technological expenditure	10,000 yuan
IV	Natural exogenic factor	*Alt_Std*	Standard error of cities' geographic altitude	–
Mediating Variables	Industrial structure	*lnRIS*	Calculated according to formula (1)	–
	Digital economy	*Digit_Eco*	Comprehensive index of digital economy	–
	Digital innovation	*Digit_Pat*	Logarithm of patent application number in digital industries	1,000 pieces

filled in using restricted linear interpolation. The cleaned data were statistically free of excessive numerical gaps, while the numerical performance of the variables was consistent with their economic significance.

3.4. Model design

Given that each year could show different developing trends, it was necessary to add time fixed effects to control for characteristic factors that changed with the year but not with individual. Because endowment characteristics, economic conditions, and carbon-neutral processes differed, it was necessary to add urban fixed effects to control for urban characteristics that did not change over time. To ensure proper model selection, this paper ran Hausman tests for fixed effect (FE) and corresponding random effect (RE) models (see Table 4).

The results of the Hausman tests indicate that differences between the FE and RE models were significant in most circumstances; hence, FE estimation was more efficient in this study. Thus, the following two-way fixed effect regression model was constructed:

Table 3
Descriptive statistics of the main variables.

Variables	Obs	Mean	Std.dev.	Min	Max
did	3832	0.165	0.372	0.000	1.000
CEE	3832	0.812	0.235	0.194	1.000
CII	3832	3.352	15.26	0.000	430.7
CNI	3832	4.792	1.731	1.099	9.138
Trd_Ind	3832	40.98	9.949	19.72	70.22
Trd_Pop	3832	54.86	13.95	21.41	87.18
SO_2	3832	1.055	1.485	−2.984	4.236
FDI	3832	10.08	1.828	4.984	13.93
Gas	3832	7.739	1.751	2.079	11.66
GDPr	3832	9.270	4.210	−3.600	19.50
Pop	3832	5.883	0.675	3.829	7.218
CO_2	3832	8.036	0.563	6.337	11.61
Gov_Out	3832	14.72	0.829	12.80	17.30
Gov_In	3832	13.81	1.115	11.35	16.95
Edu_Out	3832	12.97	0.841	10.98	15.38
Alt_Std	3734	114.4	198.5	0.577	2524
lnRIS	3832	1.661	0.661	−2.465	3.425
Digit_Eco	3832	0.320	0.112	0.061	0.802
Digit_Pat	3832	4.279	2.146	0.000	9.692

Table 4
Hausman test results.

	CEE		CII		CNI	
Prob > chi2	0.381	0.000	0.081	0.000	0.005	0.000
Controls	NO	YES	NO	YES	NO	YES
H0: Difference in coefficients not systematic.						
Test of H0	True	False	False	False	False	False

$$Car\_Neu_{it} = \alpha_0 + \beta \times did_{it} + \mathbf{Controls}_{it}\gamma + \delta_t + \mu_i + \varepsilon_{it} \tag{2}$$

In Eq. (2), $Car\_Neu_{i,t}$ stands for three dependent variables related to carbon neutralization: *CEE*, *CII*, and *CNI*. As the coefficient of the core explanatory variable, β indicates the treatment effect of policy impact. If it is significantly positive, then digital infrastructure can significantly improve the carbon neutralization of cities. *Controls* represents control variables, including proportion of the third industry, proportion of employees in the third industry, sulfur dioxide emissions, FDI, natural gas consumption, gross domestic production growth rate, and permanent population. Finally, δ_t and μ_i represent time and city fixed effects, respectively, and α_0 and ε_{it} denote the constant term and the error term, respectively.

This section is constructed on the basis of Hypotheses 3 and 4. To construct the first stage of the mediating effect model, the logarithm of industrial structure rationalization (*lnRIS*), digital economy development index (*Digit_Eco*), and digital-economy-related patents (*Digit_Pat*) were selected as dependent variables to examine whether digital infrastructure can significantly improve industrial rationalization and comprehensive digital development.

$$Med\_Var_{it} = \alpha_0 + \beta_1 \times did_{it} + \mathbf{Controls}_{it}\gamma + \delta_t + \mu_i + \varepsilon_{it} \tag{3}$$

where Med_Var_{it} includes the three mediators mentioned above. β_1 is the coefficient of interest in this model. When it is significantly positive, it shows that digital infrastructure improves industrial rationalization, digital economy development, and digital innovation. The control variables used in Model 3 were the same as in the corresponding benchmark model.

The second step was to put the core explanatory and mediating variables together to investigate whether the mediation effect exists—that is, whether digital infrastructure promotes carbon neutrality through industrial structure rationalization and innovative development in the digital economy. The model was established as follows:

$$Car\_Neu_{it} = \alpha_0 + \beta_2 \times did_{it} + \beta_3 Med\_Var_{it} + \mathrm{Controls}_{it}\gamma + \delta_t + \mu_i + \varepsilon_{it} \tag{4}$$

where Car_Neu_{it} represents the three dependent variables in the benchmark regression, while Med_Var_{it} represents the three mediating variables mentioned above. β_2 and β_3 are both coefficients of concern in this section; when significantly positive, they indicate that the mediation effect exists. The control variables used in Model 4 were the same as in the corresponding benchmark model.

4. Empirical tests and analysis

This study evaluates the effects of the Broadband China pilot policy on carbon neutrality using a two-way fixed effect (TWFE) model for DID regression. The DID design must meet several prerequisites. First, the time trend of differences in carbon neutrality indicators did not change significantly between the two groups of cities before the policy; that is, the parallel trend hypothesis was established. Second, a placebo test was required to rule out chance events. Third, since non-randomization may have produced bias, it was necessary to pay attention to the sample balance between the treatment group and the control group and use propensity score matching (PSM) and coarser exact matching (CEM) to ensure the relative balance and comparability of the two groups. In addition, possible endogeneity needs to be checked via regression of the instrumental variable (IV). Finally, since the Broadband China policy was implemented in three batches, the traditional TWFE estimators may be biased, which needs to examine the framework of staggered DID. In this study, Bacon decomposition was used to determine the severity of the bias, and a heterogeneous robust CS-DID estimator was used to correct the TWFE estimator.

4.1. Impact of digital infrastructure on carbon neutralization

The two-way fixed effects regressions with *CEE*, *CII*, and *CNI* as dependent variables, as well as the interaction term of the policy (*did*) as the core explanatory variable, were conducted according to Model 1. The results are shown in Table 5. It can be seen that the coefficients of policy intervention are significantly positive in all columns at the 1 % level, indicating that digital infrastructure has a significant role in promoting *CEE, CII,* and *CNI*.

Digital infrastructure improves production technology and management efficiency, including the use of carbon resources, by optimizing and increasing the efficiency of information transfer between production sectors. The results are presented in columns (1) and (2) of Table 5. The coefficients of *did* in these columns are significantly positive. In addition, digital infrastructure makes it possible to digitalize the management and part of the production process in the high-carbon industry. Digital transformation of the high-carbon industry is also an innovative process; switching to a digital production and management mode must be supported by many innovations. The gradual realization of digitalization will also provide resources for innovation and ultimately realize the significant progress of innovation in high-carbon industries, which is represented by the significant positive coefficients of *did* in columns (3) and (4) of Table 5. Digital infrastructure promotes the transmission and utilization of energy information while saving other aspects of operating costs, which also frees up resource space for carbon-neutral technology innovation, while digital resources and software facilities provide new carbon-neutral technology opportunities; this is reflected in the coefficients of the regression of *did* on *CNI*, which is significantly positive.

The benchmark results indicate that regional digital infrastructure can help advance carbon neutrality. Digital infrastructure has promoted the improvement of carbon efficiency, enabled the innovation of carbon-intensive industries, and provided new opportunities for the innovation of carbon-neutral technology. Technological innovation, structural changes, and efficiency changes resulting from digital infrastructure are driving carbon neutrality across several regions.

4.2. Empirical results of validity tests

(1) Parallel trend test

The DID method in the benchmark regression is based on the division of the experimental and control groups. If the two samples show a significant difference in development trends before the policy, then it does not constitute a meaningful control experiment for deriving a valid treatment effect. Therefore, it is necessary to verify that there were no significant differences between the overall development trends of the samples in the experimental and control groups before the policy. Fig. 2 shows parallel trend tests for each of the three indexes of carbon neutrality.

Although there are three variables in the measurement of carbon neutrality, the consistency of the results indicates that the process of carbon neutrality before the policy shock satisfies the parallel trend hypothesis. As shown in Fig. 2, intervention effects before the policy shocks are not significant regardless of whether control variables are implemented. In addition, a significant intervention effect was found, suggesting that regional digital infrastructure could help advance carbon neutrality.

(2) Placebo test

To verify the veracity of the treatment effects, the possibility of differential nonrandomized impact time points needs to be further ruled out; i.e., the potential impacts of major events and time trends in the year of the policy shock on the dependent variables need to be clarified. In the subsequent placebo tests, the pilot cities were randomized 500 times, and then a mixed placebo test, which suits staggered DID better, was adopted to test whether there is an effect of different cities' idiosyncrasies or specific time trends on the results.

The results in Fig. 3 show that the fake treatment effects obtained after randomizing the pilot cities roughly follow a standard normal distribution, and the true effects derived from the benchmark regression lie at the extreme end of each distribution. This indicates that, after randomization, the pilot policy no longer has a significant effect on the

Table 5
Benchmark regression results.

Variables	*CEE*		*CII*		*CNI*	
	(1)	(2)	(3)	(4)	(5)	(6)
did	0.035***	0.034***	6.409***	5.815***	1.127***	1.027***
	(0.012)	(0.012)	(1.888)	(1.609)	(0.260)	(0.227)
Controls	NO	YES	NO	YES	NO	YES
City FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
N	3832	3832	3832	3832	3832	3832
R^2	0.026	0.047	0.097	0.151	0.164	0.218

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote robust standard errors.

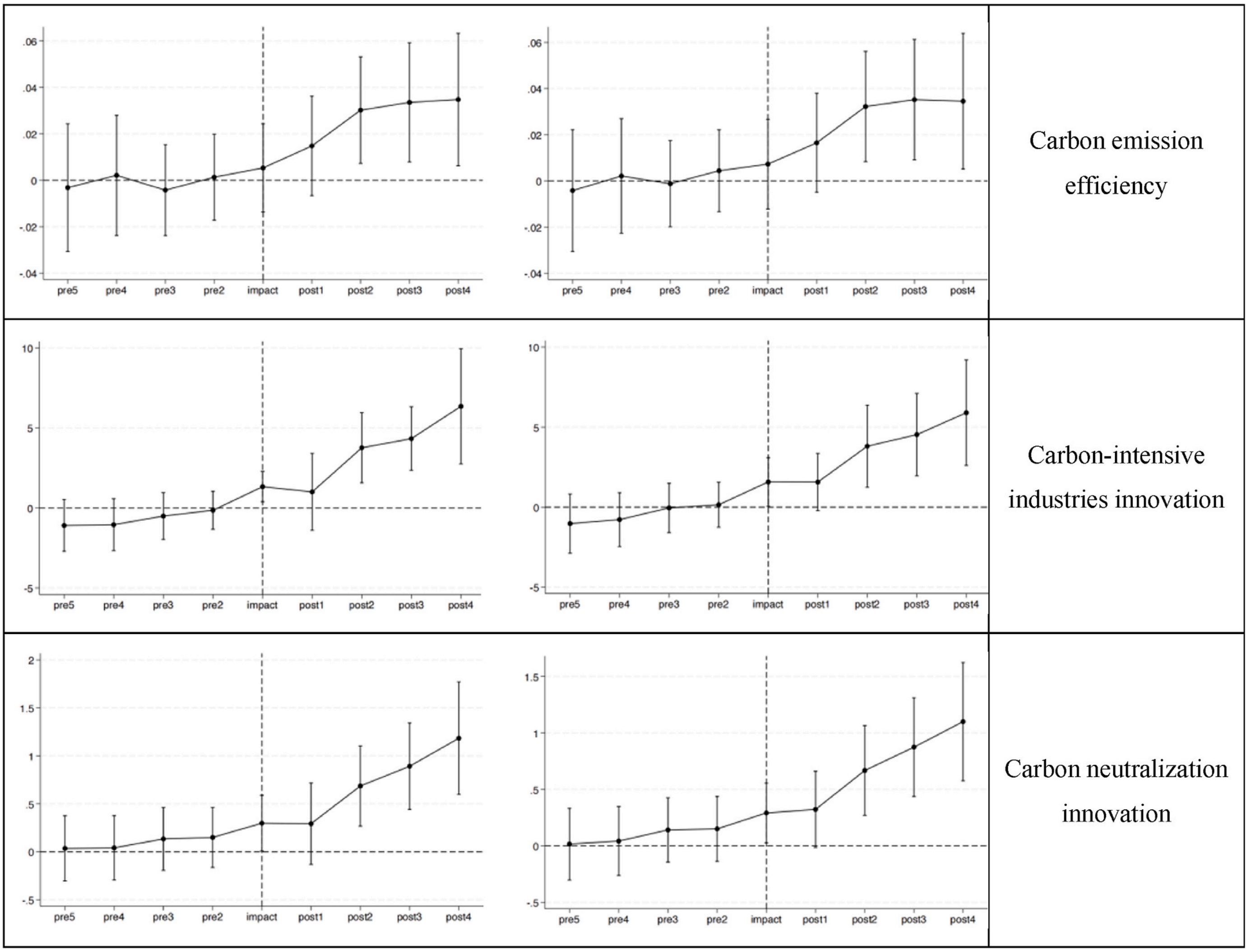

Fig. 2. Parallel trend test with and without controls.

three dependent variables, implying that the treatment effect from the benchmark regression is real and effective.

(3) Sample selection

Although the parallel trend test has verified that there is no significant difference in the development trend between the experimental and control samples before the policy shock, to ensure the robustness of the regression results and the purity of the treatment effect estimates, this section further uses PSM to remove the possibility of selectivity bias, omitted variable bias, and data bias. Table 6 shows the results using neighbor matching, kernel matching, and radius matching.

After matching, the policy variable *did* was obviously more significant in all columns but was accompanied by a slight shrinkage of the coefficients. This suggests that the original differences between the experimental and control groups, while not significant, interfered to some extent, causing a slight overestimation of the coefficients in the ordinary panel two-way fixed effects model. Hence, the results of the PSM support, in a theoretical sense, the basic conclusions drawn from the benchmark regression analysis, again indicating that the results are robust.

Although the purpose of the matching method is to screen the experimental and control group samples to improve balance, conventional PSM may fail to ensure improved variable balance after matching, while CEM can make the distribution of covariates in the experimental and control group samples as balanced as possible by observing the interference of data confounders, which ensures the improvement of covariate balance, improves the comparability of the two samples, and enhances the validity of the matching.

Table 7 demonstrates the CEM results. It can be seen that the coefficients of the core explanatory variables and their significance were tightened after forcing up the covariate balance. Due to the stricter matching mechanism, the number of samples excluded from the common range is higher, but the decidable coefficients of the model are greatly improved; all of them are more than 70 %, which indicates that the explanatory ability of the model is greatly improved compared to the benchmark regression and PSM. The coefficients of the *did* policy variable remained significantly positive at the 5 % level across all columns in Table 7, supporting Hypothesis 1 and again indicating that the regression results are robust.

(4) Endogeneity problem

This study also faces the problem of the selection of pilot cities that may not satisfy the randomization condition, and the government tends to determine the list of pilot cities based on each city's relevant conditions and development status, which can create endogeneity. Hence, this study uses the standard deviation of the elevation of cities as an instrumental variable to solve the potential endogeneity problem.

First, the standard deviation of elevation measures the magnitude of

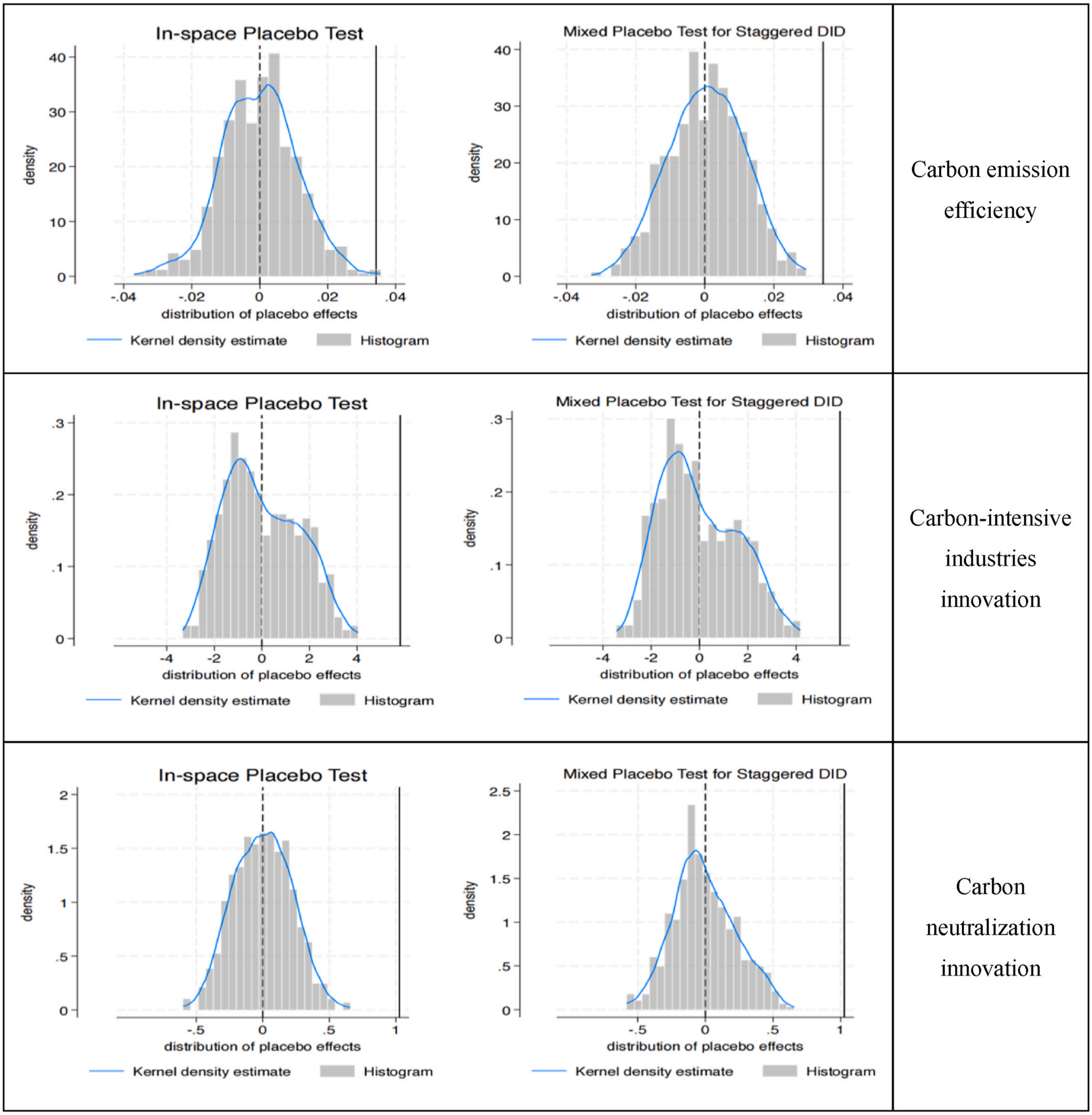

Fig. 3. In-space and mixed placebo tests.

Table 6
Results of PSM-DID estimators.

Variables	Neighbor Matching			Kernel Matching			Radius Matching		
	CEE	CII	CNI	CEE	CII	CNI	CEE	CII	CNI
did	0.039***	5.574***	0.902***	0.034***	5.690***	0.936***	0.034***	5.690***	0.936***
	(0.014)	(1.948)	(0.232)	(0.012)	(1.668)	(0.207)	(0.012)	(1.668)	(0.207)
Controls	YES	YES	YES	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
N	2684	2684	2684	3823	3823	3823	3823	3823	3823
R^2	0.054	0.183	0.262	0.046	0.139	0.206	0.046	0.139	0.206

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

Table 7
Results of the CEM–DID estimators.

Variables	*CEE*			*CII*			*CNI*		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
did	0.020**	0.025***	0.022**	4.056***	3.665***	4.034***	0.793***	0.669***	0.798***
	(0.010)	(0.008)	(0.009)	(0.687)	(0.514)	(0.600)	(0.167)	(0.095)	(0.131)
Controls	NO	YES	YES	NO	YES	YES	NO	YES	YES
City FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	YES	YES	NO	YES	YES	NO	YES
N	2244	2244	2244	2244	2244	2244	2244	2244	2244
R^2	0.880	0.882	0.884	0.803	0.817	0.823	0.727	0.758	0.776

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

the difference between cities' the topographic reliefs, which directly affect telecommunication base stations, fiber-optic cable lines, economic patterns, network costs, and other factors, thus affecting the construction and utility of the network infrastructure and meeting the correlation assumption of instrumental variables. Second, the standard deviation of elevation, as a geographic feature of a city, is not included in the selection criteria of pilot cities and is not directly related to technical indicators such as *CEE* and green innovation, thus satisfying the hypothesis of exogeneity of instrumental variables. Finally, since the elevation data of cities do not change over time within the observed period and cannot be used directly for the cross-section data of the panel regression, the interaction term of the standard deviation and the time of the cross-section elevation was constructed as the instrumental variable. The results of the instrumental variable method are shown in Table 8.

The results show that in the first-stage regression, the coefficient of the instrumental variable on the Broadband China policy is significantly negative, which confirms that too much altitude difference will impede the implementation of the network infrastructure construction and its utility, and at the same time, the F-stat is as high as 77.08, which rejects the possibility of weak instrumental variables. In the second stage, the coefficient of the policy dummy interaction term is significantly larger than the baseline regression, indicating that the positive effect of digital infrastructure on carbon neutrality still holds after solving the endogenous disturbances. It can be concluded that digital infrastructure makes a robust contribution to the carbon neutrality.

(5) Heterogeneous treatment effects problem

The Broadband China pilot policy was announced in three batches, and the time differences in policy shocks led to temporal heterogeneity of policy effects. First, the differencing process may have compared the later-treated group with the earlier-treated group, which may have resulted in a biased estimation. Second, the TWFE model in multi-period DID estimation may have assigned negative weights to the treatment effect of a certain batch of cities, thus biasing the average treatment effects. This study supplemented a Bacon decomposition for staggered DID to identify estimation biases caused by these "bad" comparisons as shown in Table 9. It shows that the sample in the "Treated later vs. earlier" group accounts for only 3.1 % of the total sample, which is a small percentage. However, this group causes reverse interference to the traditional TWFE estimator.

Table 9
Results of Bacon decomposition.

ATET decomposition	ATET component			(4) Weight
	(1) *CEE*	(2) *CII*	(3) *CNI*	
Treated vs never treated	0.040	4.570	0.837	0.926
Treated earlier vs later	0.008	2.729	0.442	0.043
Treated later vs earlier	−0.002	−4.358	−0.953	0.031

To verify robustness, heterogeneous robust CS-DID estimators were performed for each target variable using simple weighting, calendar weighting, and group weighting, respectively. The estimated results are shown in Table 10. The average treatment effects of the treated group (ATT) obtained from the heterogeneity-robust estimators still support the validity of Hypothesis 1. It can confidently be concluded that digital infrastructure can have a significant and stable promoting effect on the carbon neutralization process.

4.3. Empirical results of the mediating effect analysis

This study discusses the mechanism of digital infrastructure in facilitating carbon neutrality using mediation effect models. The *lnRIS* was used as a mediating variable, and the empirical results are shown in Table 11. It shows that digital infrastructure can effectively improve industrial rationalization conditions and work together with the rationalization of industrial structure to promote the improvement of the three indicators of carbon neutrality. The results in Table 11 confirm Hypothesis 2—that digital infrastructure promotes carbon neutralization by rationalizing industrial structure. A study that focused on digitalization produced a similar result, finding that green technology leads to industrial structure rationalization and, eventually, green growth [47].

The Digital Economy Development Index (*Digit_Eco*) and the Digital Economy patent (*Digit_Pat*) were also selected as mediators, and the

Table 8
Results of the instrumental variable method.

Variables	First step	Second step					
	did	CEE		CII		CNI	
did		0.094*	0.301***	2.800***	2.452***	4.882***	6.717***
		(0.055)	(0.075)	(0.517)	(0.647)	(0.751)	(0.928)
IV	−0.353**						
F-Stat	77.08						
Controls	NO	NO	YES	NO	YES	NO	YES
City FE	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	NO	NO	NO	NO	YES	YES
N	3734	3734	3734	3734	3734	3734	3734
R^2	0.341	0.020	0.038	0.075	0.080	0.134	0.161

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

Table 10
Results of the CS-DID estimators.

CS Estimators	Simple weighting			Calendar weighting			Group weighting		
	(1)*CEE*	(2)*CII*	(3)*CNI*	(4)*CEE*	(5)*CII*	(6)*CNI*	(7)*CEE*	(8)*CII*	(9)*CNI*
ATT	0.034**	3.450***	0.6881***						
	(0.015)	(0.935)	(0.184)						
CAveg.				0.030**	3.165***	0.619***			
				(0.014)	(0.867)	(0.166)			
GAveg.							0.034**	3.160***	0.632***
							(0.015)	(0.796)	(0.162)

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

Table 11
Mediating effect of industrial structure rationalization.

Variables	*lnRIS*		*CEE*		*CII*		*CNI*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
did	0.087***	0.113***	0.034***	0.035***	6.324***	6.321***	1.120***	1.111***
	(0.031)	(0.029)	(0.006)	(0.006)	(0.606)	(0.605)	(0.088)	(0.087)
lnRIS			0.004	0.001	0.984***	0.715**	0.076	0.026
			(0.003)	(0.004)	(0.326)	(0.334)	(0.047)	(0.051)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
City FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
N	3832	3832	3832	3832	3832	3832	3832	3832
R^2	0.091	0.233	0.026	0.031	0.099	0.103	0.165	0.188

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

empirical results are shown in Tables 12 and 13. The coefficients of *did* in columns (1) and (2) of Table 12 indicate that Broadband China policy has significantly enhanced digital economy development in pilot cities, confirming the intuition that regional digital infrastructure leads to the development of the digital economy. Other columns in Table 12 demonstrate that the digital economy does play a mediating role in promoting carbon neutrality [48]. Although the carbon efficiency change is not supported by empirical evidence, the two indicators of *CII* and *CNI* have improved significantly. The digital economy pushes the technological frontier in the field of carbon neutrality, which makes the efficiency indicators have not been significantly improved. There is no doubt that these results confirm Hypothesis 3—that digital economy development is the intermediary mechanism through which digital infrastructure affects carbon neutrality.

The results in Table 13 show that digital infrastructure has significantly facilitated digital technology innovation, confirming that the construction of digital infrastructure improves information circulation and application. In addition, digital innovation shows significant promoting effects on the *CEE*, *CII*, and *CNI*, indicating that digital-economy-related innovation plays a mediating role in the decarbonization benefits brought by digitalization. Thus, the mediating mechanism of digital technology innovation has been demonstrated, and Hypothesis 4 is accepted.

4.4. Empirical results of the heterogeneity analysis

The construction of digital infrastructures requires large up-front investments and high long-term maintenance costs after the initial construction, and network infrastructures, which are characterized by sophisticated equipment, are difficult to operate properly if they are not adequately and comprehensively maintained. Therefore, the economic development of a city determines the scale of funds that can be invested in the construction and maintenance of digital infrastructure, which in turn affects the comprehensive utility of digital infrastructure in local areas. In this section, regressions are conducted according to the median gross city product, and the results of the regressions are shown in Table 14.

It concludes that in cities with weaker economic capacity, digital infrastructure fails to exert a significant impact on carbon neutrality, while in cities with stronger economic capacity, it shows a significant uplifting effect. This heterogeneity is partly due to the fact that economically developed cities are more capable of adapting to and adopting digital infrastructure and promoting the integration of digital infrastructure construction with the existing economy with financial support. In addition, economically developed cities tend to have well-developed, mature industries and have a stronger awareness of low-carbon development, thus enabling them to quickly gain green

Table 12
Mediating effect of the digital economy.

Variables	*Dig_Eco*		*CEE*		*CII*		*CNI*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
did	0.003*	0.003**	0.035***	0.036***	0.623***	0.616***	0.109***	0.107***
	(0.002)	(0.002)	(0.012)	(0.012)	(0.060)	(0.060)	(0.024)	(0.023)
Digit _Eco			−0.105	−0.123	5.879***	6.101***	1.085***	1.056***
			(0.099)	(0.100)	(0.589)	(0.593)	(0.369)	(0.385)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
City FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
N	3832	3832	3832	3832	3832	3832	3832	3832
R^2	0.690	0.706	0.027	0.032	0.121	0.128	0.202	0.223

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

Table 13
Mediating effect of digital innovation.

Variables	*Dig_Pat*		*CEE*		*CII*		*CNI*	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
did	2.211***	2.140***	0.030**	0.031**	1.524**	1.106*	0.157**	0.113*
	(0.460)	(0.426)	(0.013)	(0.0122)	(0.758)	(0.567)	(0.068)	(0.067)
Digit _Pat			0.002*	0.001	2.210***	2.279***	0.439***	0.439***
			(0.001)	(0.001)	(0.482)	(0.469)	(0.008)	(0.008)
Controls	NO	YES	NO	YES	NO	YES	NO	YES
City FE	YES	YES	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES	YES	YES
N	3832	3832	3832	3832	3832	3832	3832	3832
R^2	0.211	0.282	0.028	0.048	0.313	0.357	0.540	0.557

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

Table 14
Heterogeneity of economic development.

Variables	Underdeveloped area			Developed area		
	(1)*CEE*	(2)*CII*	(3)*CNI*	(4)*CEE*	(5)*CII*	(6)*CNI*
did	0.026	0.326	0.009	0.025*	6.098***	1.092***
	(0.025)	(0.221)	(0.012)	(0.013)	(1.969)	(0.272)
Controls	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
N	1916	1916	1916	1916	1916	1916
R^2	0.063	0.314	0.480	0.052	0.269	0.356

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

development momentum with the support of digital infrastructure.

This study classifies 93 cities as resource-based cities and the rest as non-resource-based cities based on the National Sustainable Development Plan for Resource-Based Cities (2013–2020) issued by the State Council of China. The results of the grouping regression for cities with heterogeneous resource endowments are shown in Table 15.

It can be seen that *CEE* and *CNI* remain significant in non-resource-based cities but not in resource-based cities, while the significance of *CII* is the opposite. Many documents have confirmed that the resource curse still applies to China [49]. Qian et al. [50] found that China's resource curse is mainly transmitted through the Dutch disease effect (structural deformities in the economy), institutional effects, the crowding-out effect, and natural location and transportation factors. In terms of carbon neutrality, as resource-based cities favor low-tech traditional resource sectors, resulting in the shrinking of the manufacturing sector, the applicability of network infrastructure is low, and it is difficult for digital infrastructure to play a role in the innovation sector, which is manifested in the insensitivity of the digital infrastructure construction policy to *CEE* and *CNI*. Meanwhile, innovation in carbon-intensive industries is due to the high proportion of carbon-intensive industries in resource-based cities in which the crowding-out effect makes all aspects of capital investment mainly flow into carbon-intensive industries while other industries lack financial support. As a result, the application of digital infrastructure construction carbon-intensive industries can produce a more significant effect, making the heterogeneity unobvious under this indicator.

An environmental regulation intensity index was constructed based on the frequency of terms related to the environment in the work reports of local governments. Group regressions were conducted according to environmental regulation intensity, and Table 16 shows the regression results.

The results show that the effect of digital infrastructure in promoting carbon neutrality is larger in cities with weaker environmental regulations. According to the Porter hypothesis, technological innovation has a threshold of high up-front investment, producing a U-curve nexus between environmental regulation with technological innovation and productivity. This resulted in a lag effect in the boosting impact of environmental regulation on regional green growth [51]. Environmental regulation is likely to weaken regional innovative capability at an early stage when facing the shock of digital transformation. Regional innovation ecosystem affects green development more directly, as a main component, local government support plays an important role in infrastructure-driven tech innovation [52]. Excessive environmental regulation often distracts local governments' policy focus, hamper the integration of local innovation entities, and then slow down the whole innovation ecosystem. In summary, the carbon-neutral effect brought about by digital infrastructure is impacted by the heterogeneity of environmental regulation intensity. Nevertheless, regions with different levels of environmental regulation benefit from digital infrastructure.

5. Conclusion and implications

5.1. Conclusion

The results confirm that infrastructure-driven carbon neutrality in the digital age, including indicators such as *CNI*, *CII*, and *CEE*, has significantly improved after the implementation of policy interventions. In fact, digitalization and intelligence are driving changes in the energy supply system in the form of digital infrastructure, promoting carbon-neutral technological innovation. In addition, digitalization and intelligence are also key drivers of the transformation and upgrading of the

Table 15
Heterogeneity of resource endowments.

Variables	Other cities			Resource-based cities		
	(1)*CEE*	(2)*CII*	(3)*CNI*	(4)*CEE*	(5)*CII*	(6)*CNI*
did	0.052***	7.490***	1.272***	0.008	0.935**	0.073*
	(0.013)	(2.298)	(0.293)	(0.020)	(0.359)	(0.043)
Controls	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
N	2278	2278	2278	1554	1554	1554
R^2	0.080	0.214	0.300	0.074	0.363	0.396

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

Table 16
Heterogeneity of environmental regulation.

Variables	Weaker Environmental Regulation			Stronger Environmental Regulation		
	(1)*CEE*	(2)*CII*	(3)*CNI*	(4)*CEE*	(5)*CII*	(6)*CNI*
did	0.037***	5.116***	1.063***	0.028*	5.942*	0.828**
	(0.009)	(0.905)	(0.136)	(0.016)	(2.345)	(0.252)
Controls	YES	YES	YES	YES	YES	YES
City FE	YES	YES	YES	YES	YES	YES
Year FE	YES	YES	YES	YES	YES	YES
N	2044	2044	2044	1788	1788	1788
R^2	0.066	0.198	0.259	0.049	0.181	0.250

Note: *, **, *** denote significance at the 10 %, 5 %, and 1 % levels; values in parentheses denote standard errors.

energy-efficiency sector, which is reflected in the innovation and carbon efficiency of energy-intensive industries resulting from regional digital infrastructure.

In the mechanism tests, this study adopted industrial rationalization, the digital economy, and digital innovation to construct mediation effect models and determine the existence of these three channels. This not only affirms the importance of industrial structural changes in the digital age but also clarifies that digital economy development and digital innovation are key drivers. Although the effect of the digital economy and digital innovation driving *CEE* improvement is not fully supported by the evidence, this result is derived from the continuous advancement of cutting-edge technologies, which makes it realistic and does not contradict the hypotheses in this study. Regional heterogeneity effects have been validated, and digital infrastructure drives accelerated carbon neutrality processes, depending on regional economic development, endowment characteristics, and environmental policy synergies. Relevant policy formulation needs to take into account the heterogeneity of regional endowments and the design of differentiated initiatives.

5.2. Theoretical implications

It was found that the Broadband China pilot cities achieved improvements in three carbon neutrality indicators, indicating that digital infrastructure has significantly accelerated carbon neutralization. In addition, regional digital infrastructure was found to promote the rationalization of industrial structure and empower the development of the digital economy and digital innovation. These three factors have an important mediating impact on green development. Regional endowment also affects local digitalization and green transformation: Fiscal support is crucial in this transition, as resource dependence further affects sustainability through industrial structure and energy intensity; environmental regulation has a broader impact on short-term green performance; and policy ecosystems must be considered. These findings provide important theoretical insights into digital-driven sustainable development.

5.3. Practical implications

These conclusions have valuable implications for developing and emerging market countries to effectively leverage digital infrastructure and achieve sustainable development in the digital era.

(1) Developing countries should actively build digital infrastructure in the field of carbon neutrality and rely on digital infrastructure to leverage low-carbon economic and social development. They need to promote the construction of their digital infrastructure, and rely on digital infrastructure to complete the sustainable transformation of their economies. Governments also should focus on the use of digital infrastructure in areas such as improving carbon efficiency, facilitating the transformation of carbon-intensive industries, and supporting carbon-neutral technological innovation. By strengthening digital infrastructure, governments can provide critical support to industries that are making a low-carbon transition, thereby accelerating progress toward carbon neutrality.

(2) Industrial structure plays an important role in the nexus between digital infrastructure and carbon neutrality. Developing countries and emerging market countries should pay attention to the advancing and rationalizing adjustment of industrial structure, and make use of digital infrastructure to promote the efficient synergistic development of industrial sectors. Policymakers should focus on encouraging the application of digital technologies in traditional industries, facilitating the optimization of industrial structures and reducing the share of high-pollution, high-carbon sectors.

(3) All countries need to give full play to the potential for the digital economy to pursue carbon neutrality by encouraging digital technological innovation and digital economic development. Governments can encourage carbon-neutral smart solutions and digital technology innovation through tax breaks and subsidies and promote the synergistic development of the digital economy and renewable energy systems. Additionally, through fiscal incentives, tax reductions, and technical support, governments can foster the development of digital economy and digital innovation. Policymakers should also encourage the application of digital technologies in energy-consuming industries and promote the transformation and upgrading of high-energy-consuming industries.

(4) Governments should adopt differentiated development strategies based on the characteristics of regional endowments. Policies and implementation measures should be optimized according to the specific economic conditions and resource endowments of different regions. For example, this should include designing carbon-neutral support policies for resource-based cities and regions with high environmental stress. It also should encourage and support the application and innovation of smart energy technologies and smart production technologies, thereby unlocking the potential dividends of digital infrastructure in promoting carbon neutrality.

(5) Environmental policies interacted with each other, developing countries and emerging market countries, which were normally latecomers in the field of environmental protection, need to pay particular attention to the overall effectiveness of environmental policies. Policymakers should carefully design environmental regulations to strike a balance between environmental protection and digital infrastructure development. Excessive regulation may inadvertently divert financial resources and policy focus away from the construction of digital infrastructure, which is critical for achieving carbon neutrality. A more balanced approach would ensure that both environmental protection and digital infrastructure initiatives complement each other, maximizing the potential for digitalization to contribute to green development.

These policy recommendations aim to maximize the potential of digital infrastructure in driving carbon neutrality by promoting industrial restructuring, fostering innovation, and adopting region-specific approaches. In doing so, governments can create a more resilient and sustainable path towards achieving their carbon neutrality goals.

5.4. Limitations and future research directions

Though efforts were made to eliminate bias and conduct sound research, this study has some limitations. First, the latest data of some control variables selected in this study were not accessible, which limited the observation period (2008–2021), and the analysis of post-COVID recovery lacks sufficient observation samples. Second, the three indicators of carbon neutrality were studied in isolation, and synthesizing a comprehensive carbon neutrality indicator using entropy weighting or other methods may have led to some new findings. Finally, this study focused on different phases of carbon neutrality and introduced the structure and innovation perspectives, but there is no discussion on the synergistic development among the three objectives of low-carbon, zero-carbon, and negative-carbon.

Future research can adopt a new control variable set or calculation method to avoid data problems and obtain the latest samples to fit the current reality better. Certain synthesis algorithms could be used to develop a comprehensive carbon neutrality variable and evaluate carbon management completely. As for carbon neutrality, the interaction and coupling of emission reduction, green production, and pollution disposal can be further explored to outline an interactive and systematic view of carbon neutralization.

CRediT authorship contribution statement

Fengxiu Zhou: Conceptualization, Data curation, Formal analysis. **Lei Li:** Methodology, Visualization, Formal analysis, Writing – original draft, Data curation. **Huwei Wen:** Conceptualization, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Not applicable.

Data availability

Data will be made available on request.

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